

11. Unresolved Questions and Future Research

Candidate Count Model Research — Cross-Repository English Edition

Current scope: 134 unresolved questions across 13 research domains. Revised from the 1 August 2026 Japanese cross-repository catalog, including the RF4SP 50-trial Pink Turnip follow-up.

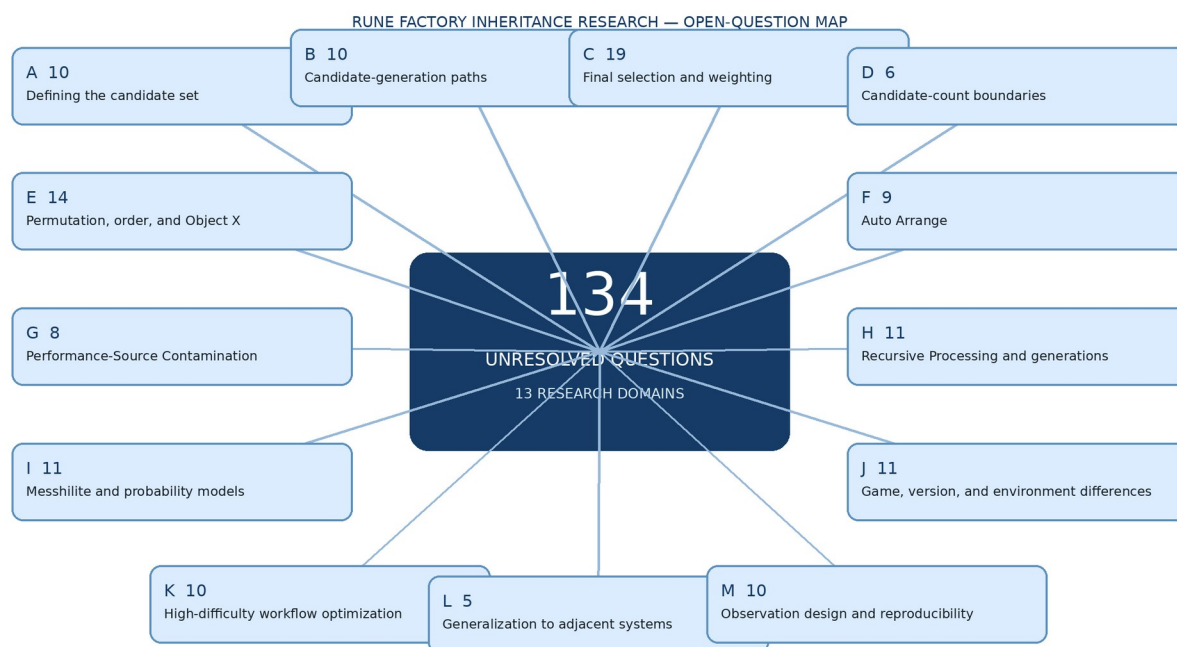


Figure 11-1. Current open-question map. Counts match the catalog below (134 total).

Purpose and authority boundary

This chapter collects unresolved questions remaining in and around the Candidate Count Model from the repository's Markdown, TXT, PDF, CSV, supplements, and observation records. It does not assert the game's internal specification. It separates questions explicitly marked unresolved from boundaries, derived questions, and interpretations placed on hold.

Status labels

Label	Meaning
Explicit	The source explicitly marks the matter unresolved, untested, unknown, or hypothesis-level.

Boundary	An observation exists, but its valid range or generalizability is not established.
Derived	The question arises when several sources are combined; it is not a direct source assertion.
On Hold	Interpretation is suspended because conditions or records were insufficient.

Important premises

- The Candidate Count Model is a replaceable approximation for organizing real-device observations, not proof of internal code.
- The 4,000 RF5 trials strongly support model compatibility for the tested three-to-six Glitter Augite Messhilite conditions, not the entire internal process of generation, weighting, ordering, or recursion.
- Under the RF4SP candidate-overflow condition, combinations excluding Pink Turnip appeared 42 times in 50 trials. This is difficult to explain with four equally likely combinations, but it does not by itself prove weighted selection or falsify the entire Candidate Count Model.
- CSV records provide observations and aggregates; they do not directly identify an internal algorithm.

A. Defining the candidate set (10)

A-01 [Explicit] What counts as one candidate?

It is unresolved whether the base item, Performance Source, internal arrangements, required materials, Auto Arrange materials, and Performance-Source Contamination should be counted as equivalent candidate units.

A-02 [Explicit] Which shoe/accessory count is valid: 4/5/6 or 5/6/7?

The count changes depending on whether Performance-Source Contamination is an independent candidate; this must be reconsidered separately from the directional claim that larger candidate pools lower success.

A-03 [Explicit] Are the Performance Source item and its internal arrangements separate candidates?

Both may become candidates simultaneously, but the identifying conditions and duplication relationship are unresolved.

A-04 [Explicit] Do repeated identical materials create separate candidates, a same-type population, or a compressed candidate?

Messhilite tests show success changing with the number of same-type inputs, but the internal representation is Unknown.

A-05 [Explicit] How are duplicate same-name or same-type candidates processed?

It is unknown whether they remain separate instances, collapse into one equivalent candidate, or merge only under some conditions.

A-06 [Explicit] Does candidate compression exist, and when does it occur?

Some results change less than the apparent candidate-count increase predicts; compression, fixed slots, and priority remain competing explanations.

A-07 [Explicit] What excludes an element from the final candidate set?

The exclusion rules and the processing stage at which exclusion occurs are unresolved.

A-08 [Explicit] What is the effective upper limit on candidate count?

The maximum remains unknown when crafting inputs, three arrangement slots, recursive expansion, and contamination are combined.

A-09 [Derived] Can candidate existence be equated with selectable probability?

If priorities or fixed slots exist, presence alone cannot express effective candidate count; an effective or weighted N may be needed.

A-10 [Derived] Is the candidate set built once or reconstructed at several stages?

A multistage path - generation, compression/priority processing, then final three-slot selection - remains possible.

B. Candidate-generation paths (10)

B-01 [Explicit] Exactly when does a directly inserted material become a candidate?

Differences by equipment type, input position, Performance Source role, and additional-material role remain unresolved.

B-02 [Explicit] Exactly when do required recipe materials become candidates?

The Auto Arrange hypothesis for recipes requiring more than three materials has not been confirmed across recipes, categories, and games.

B-03 [Explicit] Is only the Performance Source specially referenced?

Its internal data appears easier to reference than that of additional equipment, but position and processing role have not been separated.

B-04 [Explicit] When are the internal arrangements of additionally inserted equipment expanded?

It is unknown whether they are never expanded or merely absent under the tested conditions.

B-05 [Explicit] When is Auto Arrange-derived information reintroduced as later candidates?

Retention in the finished item appears plausible, but the relation between retention and re-expansion is unresolved.

B-06 [Explicit] What triggers Performance-Source Contamination?

It is observed strongly in shoes and accessories; category dependence and the behavior of weapons, armor, and shields remain unorganized.

B-07 [Explicit] How does inheritance depth change candidate count?

Candidates tend to accumulate through re-inheritance, but the increase per generation is not established.

B-08 [Derived] What happens when the same candidate is generated through multiple paths?

Required material, internal arrangement, and contamination paths may create duplication, compression, or increased weight.

B-09 [Derived] When are candidates generated?

Possible stages include crafting start, Performance Source selection, Auto Arrange, and immediately before final selection.

B-10 [Derived] How does generated candidate data relate to displayed information?

Blue text, descriptions, and finished stats may not expose all internal candidates.

C. Final selection and weighting (19)

C-01 [Explicit] Is the final selection fully uniform?

The base model assumes uniform sampling without replacement, but this is not proven as the internal implementation.

C-02 [Explicit] Does weighted selection exist?

High- and low-frequency candidates, retained arrangements, and displaced Auto Arrange candidates motivate the hypothesis.

C-03 [Explicit] What determines candidate weight?

Candidate type, input order, existing arrangements, Performance Source, equipment category, same-type count, generation, and game remain confounded.

C-04 [Explicit] Is there an internal priority order?

Existing arrangements or Performance Source data may outrank Auto Arrange data, but evidence includes very small samples and cannot support confirmation.

C-05 [Explicit] Do fixed slots exist?

A frequently retained candidate might be heavily weighted or occupy a non-lottery slot; current observations cannot distinguish them.

C-06 [Explicit] Is selection staged?

Category-then-candidate selection or retained-slot-then-gap-filling are possible multistage models.

C-07 [Explicit] How many internal draws occur?

One three-item combination, sequential draws, and re-draws after invalid candidates remain possible.

C-08 [Explicit] When the pool exceeds capacity, is there a redraw, reconstruction, or shuffle?

Display-order changes cannot uniquely identify the internal event.

C-09 [Explicit] Is selection without replacement?

The approximation assumes it, but repeated same-type inputs may still retain distinct internal identities.

C-10 [Explicit] Is there category-level priority?

Base items, materials, existing arrangements, and Auto Arrange data may be prioritized by category.

C-11 [Boundary] Under what conditions do equal N values produce different distributions?

If composition or generation path matters, the application boundary of an N-only model must be identified.

C-12 [Derived] Can weighting be statistically distinguished from compression?

Weights, fixed slots, duplicate compression, and staged selection can generate similar observed bias.

C-13 [Explicit] What are the RNG seed, consumption points, and draw timing?

Seed updates, load behavior, and random-number consumption between crafting actions are unresolved.

C-14 [Derived] Can repeated save/load trials be treated as independent?

The 4,000-trial analysis uses practical independence, but internal seed-series independence was not directly checked.

C-15 [Explicit] What causes the RF4SP Pink Turnip candidate-overflow result?

In 50 observations, combinations excluding Pink Turnip appeared 42 times (84%); chi-square 93.84, df=3, Cramer's V about 0.791, inconsistent with a simple four-combination uniform model under that condition.

C-16 [Explicit] What causes the Pink Turnip bias?

Material-specific weight, item ID or registration order, input order, generation order, legacy values, and other processing remain candidates.

C-17 [Boundary] How far does the Pink Turnip bias reproduce?

Independent replication, material substitution, position changes, and an RF5 counterpart are still needed.

C-18 [Derived] Can subset weight be distinguished from individual-candidate weight?

The 42/50 result can arise from item weight, combination weight, staged selection, fixed slots, or generation order.

C-19 [Derived] When may the uniform Candidate Count formula be applied?

The Messhilite 4,000 trials fit the approximation while the Pink Turnip condition did not; game, composition, material properties, and generation path may define the boundary.

D. Candidate-count boundaries (6)

D-01 [Explicit] Is there a real boundary near $N=3$?

$N \leq 3$ often appears near-fixed and $N \geq 4$ more competitive, but neither absolute fixation nor guaranteed collapse has been established.

D-02 [Explicit] Is the boundary total N , category count, or the number of open slots?

Occupancy of the three existing slots may matter more than raw N .

D-03 [Explicit] Is boundary behavior the same in RF4SP and RF5?

RF4SP appears random earlier; RF5 appears more fixed near $N \leq 3$, but the comparison is provisional.

D-04 [Explicit] What changes when the boundary is crossed?

Candidate generation, final selection, display order, redraw, and displacement remain unresolved alternatives.

D-05 [Boundary] Why do some Gold Bracelet cases change little as apparent N grows?

Priority, fixed slots, compression, and test conditions remain indistinguishable.

D-06 [Derived] Does the boundary differ by equipment category or recipe?

Shoe/accessory observations cannot be generalized directly to weapons, armor, and shields.

E. Permutation, order, and Object X (14)

E-01 [Explicit] Are candidate selection and arrangement order separate processes?

The desired three candidates may be selected first and ordered afterward.

E-02 [Explicit] Are all final three-slot permutations equally likely?

Combination probability can be multiplied by an order approximation, but permutation uniformity is unproven.

E-03 [Explicit] Is there a post-selection shuffle or an ordered draw?

Identical displays can result from different internal models.

E-04 [Explicit] Do input order, recipe order, or source-internal order affect final placement?

Fixed and unordered behavior coexist; the retained order signal is unresolved.

E-05 [Explicit] What is the success probability for placing Object X first?

Having the right candidates and placing Object X first are distinct requirements.

E-06 [Explicit] Does Object X have a special effect on ordering?

Its placement may use ordinary weight or a special inversion-related priority or constraint.

E-07 [Explicit] Does the roughly 54% Mealy Apple-first result reproduce?

This RF4SP 300-trial observation still requires evaluation against 50% and under other conditions.

E-08 [Explicit] Is the order bias unique to Mealy Apple?

Negative-stat materials as a class and item-specific effects remain alternatives.

E-09 [Explicit] What are the order distributions of other negative materials?

Heavy Powder, Sour Drop, Mixed Herbs, Sweet Powder, and comparable materials lack matched tests.

E-10 [Explicit] Is the list of negative materials complete?

Other materials relevant to Object X reversal may remain unidentified.

E-11 [Explicit] How does order bias relate to overflow redraw or reconstruction?

The same order rule may not operate at low N and $N \geq 4$.

E-12 [Derived] What is the general success model including order?

Combination probability, post-selection permutation, and placement weights have not been unified.

E-13 [Explicit] Are Pink Turnip selection bias and placement bias one phenomenon?

It may be selected less often, placed later after selection, or both.

E-14 [Derived] Is one priority value reused for selection and placement?

The same item value or generation order might affect both stages, or separate weights may exist.

F. Auto Arrange (9)

F-01 [Explicit] Does the n-3 trigger rule hold universally?

The required-material 4-to-1 and 5-to-2 pattern is not guaranteed across all equipment and games.

F-02 [Explicit] Are required materials sampled uniformly?

Early, middle, and late materials have appeared, but their selection rates and weights are unknown.

F-03 [Explicit] Does recipe position affect selection rate?

A last-material-only claim is disfavored, but complete position independence is unproven.

F-04 [Explicit] How are Auto Arrange candidates displaced when existing arrangements are present?

Gap filling, low priority, and exclusion remain competing models with limited trials.

F-05 [Explicit] How is Auto Arrange information stored?

Material identity, order, and generation-path metadata may be retained differently.

F-06 [Explicit] When does stored Auto Arrange data recursively expand?

The candidate-count debt interpretation has not been mapped across conditions.

F-07 [Explicit] How do RF4SP and RF5 differ in randomness?

RF4SP appears random from the first trial and RF5 more boundary-dependent, but large matched comparisons are lacking.

F-08 [Explicit] What is the priority relation between Auto Arrange and contamination?

It is unknown which survives more often and whether the rule is category-specific.

F-09 [Derived] Under what workflow can Auto Arrange be fully avoided?

Avoidance may be difficult in advanced recipes; the optimal process search is incomplete.

G. Performance-Source Contamination (8)

G-01 [Explicit] What is the formal unit and name of the phenomenon?

The Current provisional name better reflects the observed source, but the definition depends on what exactly contaminates the pool.

G-02 [Explicit] Which categories exhibit it?

It is strong in shoes and accessories; equivalent processing in weapons and other categories remains unresolved.

G-03 [Explicit] Do contamination rates differ between RF4SP and RF5?

RF5 may be higher, but controlled matched trials are insufficient.

G-04 [Explicit] Can the source item and its internal arrangements both become candidates?

Simultaneous and conditional candidate formation have not been separated.

G-05 [Explicit] How do inheritance generation, contamination rate, and N relate?

Growth is observed, but linear increase, saturation, and truncation remain possible.

G-06 [Explicit] How does contamination interact with Auto Arrange?

It may displace Auto candidates or compete within one pool.

G-07 [Explicit] Where is the boundary between contamination and Recursive Processing?

A method to assign each observed candidate to its generation path is incomplete.

G-08 [Derived] What is the actual probability of removing the Performance Source?

The five-candidate RF5 shoe example is explanatory, not an observed candidate-generation probability.

H. Recursive Processing and generations (11)

H-01 [Explicit] Exactly when does Recursive Processing occur?

It is a hypothesis compatible with observation, not a universal process confirmed for every inheritance.

H-02 [Explicit] What can recursive reference target?

Performance Sources, added equipment, Auto Arrange data, contamination-derived data, and inherited information are not fully classified.

H-03 [Explicit] Is recursion determined by position or processing role?

Observed source-side reference has not separated slot location from semantic role.

H-04 [Explicit] What is the recursion-depth limit?

The number of generations that can remain referenceable is untested.

H-05 [Explicit] What is the expansion-count limit?

Deep structures may expand fully or stop at an internal cap.

H-06 [Explicit] What is the generation cap and truncation rule?

Oldest-first removal, newest rejection, and priority selection remain possible.

H-07 [Explicit] When is internal information compressed or lost?

Unexpected retention and mid-chain loss are not explained by one model.

H-08 [Explicit] Do RF4SP and RF5 have different recursion depths?

The suggestion that RF4SP is deeper and RF5 shallower remains provisional.

H-09 [Explicit] Is order information inherited during recursive expansion?

Candidate identity and internal ordering may be retained or re-decided separately.

H-10 [Derived] How are cycles or repeated expansion prevented?

Some internal rule must prevent unlimited reappearance across generations, but it has not been observed.

H-11 [Derived] Can non-expansion be distinguished from expansion followed by compression?

Invisible deep candidates support both explanations.

I. Messhilite and probability models (11)

I-01 [Boundary] How far can the 4,000 RF5 trials be generalized?

The result covers specific materials, equipment, and 3-to-6-stack procedures; other games and inputs remain outside direct observation.

I-02 [Explicit] What internal mechanism produces the same-type population effect?

Increasing same-type inputs raises successful combinations in the approximation, but internal identity handling is unknown.

I-03 [Explicit] Have fully different and mixed same-type structures been measured at scale?

Theoretical treatment exists, but matched large trials are incomplete.

I-04 [Explicit] What is the ordered success probability when Object X is included?

A combined model for candidate success plus first-position success is unfinished.

I-05 [Explicit] What rule governs performance loss when Messhilite is selected?

Loss of one desired material effect appears plausible but is not guaranteed for every material and condition.

I-06 [Explicit] How do near-fixed behavior, priority, and redraw change theoretical probability?

Conditions that diverge from uniform combinations have not been quantified.

I-07 [On Hold] What caused the ATK +22 versus +52 difference in six-Glitter builds?

Iron versus Dragonic Stone and standalone versus inherited crafting were confounded.

I-08 [On Hold] What caused the ATK +10 / M.ATK +5 addition?

It matches a total-item-level bonus but does not appear every time; material and item levels remain untested.

I-09 [Explicit] How can Iron, Dragonic Stone, base weapon, source, and input count be separated?

A controlled experiment for I-07 has not been completed.

I-10 [On Hold] What was the historical input order?

The remembered Glitter-Glitter-Messhilite and later Messhilite-Glitter-Glitter orders lack a preserved record.

I-11 [Derived] Are candidate success and final stat determination separate models?

Successful selection may precede a separate performance-value process.

J. Game, version, and environment differences (11)

J-01 [Explicit] How does original RF4 differ from RF4SP?

Porting and fixes may have changed internal behavior; no controlled comparison exists.

J-02 [Explicit] How do candidate-count boundaries differ between RF4SP and RF5?

Only partial comparisons exist.

J-03 [Explicit] How does near-fixed behavior differ?

Current descriptions remain provisional and need matched conditions.

J-04 [Explicit] At what point does ordering become unstable in each game?

RF5 may shift near $N \geq 4$ and RF4SP earlier, but neither boundary is confirmed.

J-05 [Explicit] Do contamination rates differ?

RF5 appears higher, but controlled evidence is lacking.

J-06 [Explicit] Does Auto Arrange priority differ?

RF5 may show stronger gap-filling behavior; this is interpretation pending verification.

J-07 [Explicit] Do compression, priority, and weighting differ?

The two games may implement materially different selection structures.

J-08 [Explicit] Does recursion depth differ?

No cross-game depth study has been completed.

J-09 [Explicit] Does inheritance order matter differently?

RF4SP remains unresolved and RF5 only hypothesis-level.

J-10 [Explicit] Do hardware, patch, or port versions differ?

Steam/Switch versions, patches, and reproducible environment metadata are not organized.

J-11 [Derived] Can the model generalize to RF3 and other titles?

Candidate selection in material shops and adjacent systems is untested.

K. High-difficulty workflow optimization (10)

K-01 [Explicit] What is the optimal path to one target result?

Stage splitting, intermediate items, and candidate control have not been exhaustively searched.

K-02 [Explicit] What should optimization minimize?

Expected attempts, material use, inspection work, reloads, and mental burden conflict.

K-03 [Explicit] Does ability compression truly reduce N?

Recursive re-expansion may recreate candidates after compression.

K-04 [Explicit] What is the tradeoff between avoiding Auto Arrange and avoiding contamination?

The claim that both cannot be fully avoided has not been tested per equipment category.

K-05 [Explicit] Does the 25% Ghost/Rocket draw actually exist?

It is an explanatory value under a four-candidate assumption, not a directly observed Current success rate.

K-06 [Explicit] What is the RF5 Fairy Boots source-removal probability?

The five-candidate example does not establish the generation process or actual rate.

K-07 [Explicit] What is the minimum number of 25% stages?

Reducing three stages to one appears difficult and to two plausible; the search is incomplete.

K-08 [Explicit] How much work does split-saving actually save?

It reduces rollback scope rather than probability, but no quantitative comparison exists.

K-09 [Derived] Where do theoretical and practical optima diverge?

The exchange rate between one extra ability and attempts, materials, and time is undefined.

K-10 [Derived] Can a workflow-optimization algorithm be built?

Unknown generation, recursion, weighting, and order prevent an exact path-search model.

L. Generalization to adjacent systems (5)

L-01 [Explicit] Do shops and traded goods use weighted selection?

Heinz, Leon, Raven, Sakuya, Terry, and similar systems remain hypothesis-level analogies.

L-02 [Explicit] How does the shipped-item pool affect storefront appearance rates?

Quantitative testing of pool dilution is insufficient.

L-03 [Explicit] Are shop candidate pools shared or separated by category?

Crops, flowers, materials, and traded goods may use distinct pools.

L-04 [Explicit] How do storefront slots handle duplicates and selection?

Without-replacement, weighted, and staged selection remain unresolved.

L-05 [Derived] Is the inheritance model truly the same as the shop-pool problem?

Structural similarity does not justify using the same equation or constraints.

M. Observation design and reproducibility (10)

M-01 [Explicit] How many trials are needed to detect low-frequency bias?

No power design exists for weighting, order bias, and game differences.

M-02 [Explicit] Which conditions still need simultaneous control?

Equipment, material level, item level, input order, Performance Source, internal arrangements, and patch environment are often not all fixed.

M-03 [Derived] How can unobservable internal candidates be inferred?

Finished results cannot uniquely recover generation, selection, and ordering.

M-04 [Derived] What is the smallest experiment that separates competing hypotheses?

Uniform selection, weighting, fixed slots, compression, and staged selection need discriminating controls.

M-05 [Explicit] Which historical observations are irrecoverable?

Some input orders and conditions remain memory-only and cannot be reanalyzed.

M-06 [Boundary] What does agreement within a 95% confidence interval establish?

It shows compatibility, not exclusion of competing models with the same predictions.

M-07 [Derived] How should multiple comparisons and exploratory observation be controlled?

Searching many materials, orders, and games raises the chance of incidental bias.

M-08 [Derived] How complete is public raw-data coverage?

The 4,000-trial CSV is strong, but order-bias and boundary-test logs may not have equal coverage.

M-09 [Explicit] What follow-up design can identify the Pink Turnip cause?

Material substitution, position swaps, generation-order swaps, fixed N, and fixed recipe/equipment should be combined.

M-10 [Derived] How should post-hoc discovery affect the statistical interpretation?

Because 20 trials followed an initial 30-trial bias, the confirmation subset may need separate reporting from the full 50.

Core research bundles

1. Defining N

A-01–A-10 must be resolved before N can carry one stable meaning across conditions.

2. Uniform selection versus weighting, fixed slots, compression, or staged selection

C-01–C-19 form an identification problem among competing mechanisms. The 50-trial Pink Turnip observation supplies a concrete condition where uniformity cannot be assumed automatically.

3. Separating combinations from permutations

E separates the probability of obtaining the desired set from the probability of placing it in the required order.

4. Separating recursion from contamination

G and H require attribution of each candidate to its generation path.

5. Separating shared structure from game-specific behavior

J is required before RF4SP and RF5 can be described by one shared core plus title-specific layers.

6. Moving from model fit to mechanism identification

The 4,000 trials support fit under tested conditions; the next stage is observation design that distinguishes mechanisms making the same prediction.

Primary source media

Markdown: README.md; ROADMAP.md; Candidate-Count-Model.md; Success-Probability.md; Auto-Arrange.md; Self-Contamination.md; Recursive-Processing.md; Messhilitite-Inheritance.md; The-Hidden-Cost-of-Shipping.md.

PDF: Chapters 03–12 and research/01–06. TXT: additional unresolved-question memo, crop/flower storefront note, Chapter 07 column, and convergence note. CSV: research/07 integrated data. Google Drive observation records: RF4SP overflow weighting, Object X/order tests, and RF4SP/RF5 within-three-slot comparisons.

Collection boundary

Repository duplicates were not counted twice. Images and Mermaid files were used to confirm conceptual relationships, not to infer new internal specifications. Derived questions require human approval before adoption as repository claims. The Pink Turnip result reproduces a bias under one condition; its cause remains Unknown.

Translation boundary: This English edition preserves the 134 issue identities, evidence-state labels, numerical conditions, and Unknown boundaries of the Current Japanese catalog. The map, domain counts, and body were reconciled by Question ID before publication.